

KIRUNA, Sweden

WMO: 020440

Lat: 67.823N

Lon: 20.340E

Elev: 447

StdP: 96.07

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.0	(c) -26.0	(d) -32.7	(e) 0.2	(f) -29.0	(g) -29.2	(h) 0.3	(i) -25.9	(j) 13.1	(k) -2.9	(l) 11.4	(m) -4.3	(n) 1.4	(o) 220	(p) 0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.4	(c) 23.1	(d) 14.7	(e) 21.0	(f) 13.9	(g) 19.0	(h) 13.2	(i) 16.2	(j) 21.0	(k) 15.1	(l) 19.0	(m) 14.0	(n) 17.8	(o) 3.5	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	10.8	17.6	13.2	10.0	16.7	12.1	9.3	15.6	47.1	21.3	43.7	19.0	40.6	17.9	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.0	7.8	-32.2	25.4	3.2	1.8	-34.5	26.7	-36.3	27.8	-38.1	28.8	-40.5	30.2
			-32.3	17.5	3.2	1.2	-34.6	18.4	-36.5	19.1	-38.2	19.7	-40.5	20.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-12.3	-11.5	-7.6	-1.6	4.5	9.8	13.3	11.0	6.4	-0.6	-6.5	-9.4		
	DBStd	10.16	7.25	7.48	5.53	4.19	4.13	3.79	3.28	3.15	3.41	4.68	5.39	6.47		
	HDD10.0	3992	690	603	547	349	180	49	7	25	114	329	495	602		
	HDD18.3	6816	948	836	805	599	431	257	160	228	357	588	745	861		
	CDD10.0	222	0	0	0	0	8	43	109	56	7	0	0	0		
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0		
	CDH23.3	47	0	0	0	0	2	8	33	4	0	0	0	0		
	CDH26.7	2	0	0	0	0	0	1	2	0	0	0	0	0		
(14) Wind	WSAvg	3.3	3.0	3.4	3.9	3.5	3.5	3.4	3.0	2.8	3.3	3.2	3.0	3.1		
(15) Precipitation	PrecAvg	502	27	23	24	27	34	54	84	73	46	40	38	32		
	PrecMax	656	71	48	55	66	93	177	150	149	84	90	91	84		
	PrecMin	346	9	5	2	3	3	7	24	15	15	8	14	12		
	PrecStd	92	14	12	14	17	20	36	36	35	20	21	17	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	5.2	5.5	11.4	22.2	24.2	26.7	23.7	17.9	10.6	6.2	5.4		
		MCWB	2.4	2.3	2.1	6.4	11.7	14.4	16.9	15.9	12.0	7.8	4.0	3.0		
	2%	DB	2.0	2.8	3.2	7.9	17.8	21.2	24.0	21.3	15.1	9.0	3.7	2.7		
		MCWB	0.1	0.7	0.8	3.8	10.1	12.7	15.1	14.9	11.0	7.0	2.3	0.8		
	5%	DB	-0.7	0.6	1.9	6.1	14.1	19.0	21.9	18.9	13.4	7.6	1.9	0.6		
		MCWB	-1.9	-1.0	-0.1	2.9	8.4	11.8	14.8	14.0	10.2	6.0	0.8	-0.6		
	10%	DB	-2.7	-1.4	0.2	4.7	11.2	16.8	19.7	16.9	12.0	5.8	0.5	-1.0		
		MCWB	-3.7	-2.4	-1.5	1.9	6.7	11.1	14.0	12.7	9.4	4.5	-0.2	-1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.8	2.4	6.5	12.5	16.2	18.2	17.6	13.2	8.6	4.2	3.2		
		MCDB	4.8	5.2	5.0	11.2	20.1	21.9	23.1	22.2	15.9	9.5	5.8	5.2		
	2%	WB	0.2	0.7	1.1	4.3	10.6	14.2	16.7	15.7	11.8	7.2	2.3	1.1		
		MCDB	1.7	2.7	3.0	7.1	16.8	18.5	21.8	19.7	14.1	8.8	3.4	2.3		
	5%	WB	-1.8	-1.0	-0.1	3.1	8.6	12.8	15.6	14.5	10.8	5.9	0.9	-0.6		
		MCDB	-0.7	0.3	1.6	5.8	13.7	17.6	20.1	18.1	12.9	7.5	1.6	0.4		
	10%	WB	-3.6	-2.6	-1.4	2.1	7.1	11.4	14.6	13.2	9.8	4.4	-0.2	-1.9		
		MCDB	-2.7	-1.6	0.2	4.5	10.7	16.0	18.5	16.3	11.6	5.5	0.4	-1.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.0	8.9	8.2	7.7	8.0	8.4	8.2	7.3	5.7	6.4	7.5		
		MCDBR	7.9	8.6	8.2	8.8	12.6	11.6	11.8	11.8	9.4	7.0	6.4	7.5		
	5% WB	MCWBR	7.1	7.2	6.7	5.9	6.9	5.5	5.3	6.4	6.0	5.2	5.6	6.5		
		MCWBR	7.8	7.6	7.8	8.1	11.4	10.1	9.9	10.5	7.9	6.5	5.8	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.169	0.230	0.269	0.297	0.317	0.322	0.328	0.315	0.284	0.252	N/A	N/A		
		taud	1.933	2.141	2.232	2.268	2.451	2.520	2.516	2.532	2.585	2.351	N/A	N/A		
	Ebn at Noon		149	598	781	856	881	885	865	840	789	581	N/A	0		
		Edh at Noon	9	45	75	97	91	88	86	76	55	37	N/A	0		
(44) All-Sky Solar Radiation	RadAvg		0.04	0.50	1.97	3.88	4.95	4.99	4.66	3.37	1.84	0.61	0.10	0.00		
	RadStd		0.00	0.04	0.10	0.17	0.42	0.39	0.45	0.28	0.22	0.09	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page